

# Project Report

## Smart City AQI Prediction System

### Introduction

India faces a severe urban air quality crisis that extends beyond Delhi to metros like Mumbai, Kolkata, Bengaluru, and Chennai. In 2024-25, Delhi averaged an AQI of 218, classified as 'Poor' or worse for over 200 days, while other cities also experienced hazardous pollution levels. This crisis contributes to an estimated 1.67 million premature deaths annually, disproportionately impacting urban populations. Despite deploying over 900 Continuous Ambient Air Quality Monitoring Stations (CAAQMS), actionable multi-agency response protocols are lacking, with only 31% of cities linking data to interventions. This project addresses the urgent need for an AI-powered Urban Air Quality Intelligence platform that integrates multi-modal data sources — including CAAQMS, satellite imagery, mobility feeds, meteorological forecasts, and geospatial land use layers — to provide hyperlocal predictive forecasting, geospatial pollution source attribution, and enforcement intelligence. The platform empowers smart city administrators with proactive, evidence-based tools to reduce pollution at its source and improve public health.

### Problem Statement

India's urban air quality crisis is a national challenge requiring proactive, AI-driven solutions. Current AQI monitoring systems are reactive, limited in coverage, and lack predictive and enforcement intelligence capabilities. This project aims to build an AI-powered Urban Air Quality Intelligence platform that fuses data from monitoring stations, satellite imagery, mobility feeds, meteorological forecasts, and geospatial land use layers.

Key objectives include:

- **Geospatial Pollution Source Attribution:** Multi-modal AI analysis of spatial-temporal AQI patterns against land use maps, traffic density, construction permits, industrial emissions, and satellite-detected thermal anomalies to attribute pollution sources at ward or zone level with statistical confidence.
- **Hyperlocal Predictive AQI Forecasting:** AI models integrating meteorological forecasts, traffic predictions, seasonal emission calendars, and atmospheric dispersion modeling to provide 24-72 hour AQI forecasts at 1km grid resolution, enabling scheduled interventions.
- **Enforcement Intelligence & Prioritization:** Correlation of pollution hotspot data with registered emission sources such as industries, construction sites, waste burning locations, and diesel fleet movements to generate prioritized, evidence-backed enforcement recommendations with geospatial documentation.

The solution moves beyond static dashboards to deliver actionable, evidence-based insights for smart city administrators to anticipate pollution trends, identify sources, and prioritize

enforcement actions for impactful intervention.

## Objectives

- Develop regression models to predict AQI values using historical and contextual data.
- Compare baseline linear regression with advanced ensemble methods for accuracy and interpretability.
- Deploy the most interpretable and stable model in a user-friendly Streamlit app for real-time predictions.
- Ensure reproducibility, transparency, and scalability for smart city integration and hackathon evaluation.

## Data & Features

- Core datasets: Historical AQI from CAAQMS, satellite imagery, traffic and emissions data, meteorological forecasts, and geospatial land use indicators.
- Feature engineering:
  - Rolling averages (7-day, 30-day AQI)
  - Lag features (1-day, 3-day, 7-day AQI)
  - Seasonality indicators (month, day of week)
  - Smart mobility indicators (road occupancy, emissions, sentiment, energy consumption)

## Model Development & Results

| Model             | MSE  | R <sup>2</sup> | Comments                                                                |
|-------------------|------|----------------|-------------------------------------------------------------------------|
| Linear Regression | 2591 | 0.08           | Weak baseline, underfitting.                                            |
| Ridge Regression  | 810  | 0.71           | Strong baseline, interpretable coefficients, robust across folds.       |
| Random Forest     | 805  | 0.71           | Slightly better accuracy, but less interpretable and heavier to deploy. |

- Ridge Regression selected for deployment because:
  - Comparable accuracy to Random Forest.
  - Clear coefficient interpretation, important for policy and citizen trust.
  - Lightweight, faster inference, easier integration in Streamlit.
- Streamlit app (app.py) hosts the Ridge model and scaler (.joblib), enabling real-time AQI predictions with simple input forms and visualizations.

## Expected Outcomes

- Reliable hyperlocal AQI forecasts for city wards and zones.

- Transparent, interpretable model outputs guiding actionable strategies.
- Scalable prototype ready for hackathon submission and smart city deployment.

## Future Scope

- Combined with RAG function for context aware responses.
- Integrate real-time weather and IoT sensor data for dynamic updates.
- Explore deep learning models (LSTMs, Transformers) for time-series AQI prediction.
- Develop multi-city comparative intelligence dashboards to track intervention effectiveness.
- Build citizen health risk advisory systems with multi-language support.
- Extend deployment to mobile apps and REST APIs for broader accessibility.

## Conclusion

This project presents a comprehensive, AI-powered Urban Air Quality Intelligence platform addressing India's critical urban pollution challenge. By leveraging multi-modal data fusion, hyperlocal predictive forecasting, geospatial source attribution, and enforcement intelligence, it empowers smart city administrators with proactive, evidence-based tools to reduce pollution at its source. The choice of Ridge Regression balances accuracy, interpretability, and deployment efficiency, making the solution both hackathon-ready and a foundation for real-world smart city applications.

Authors:

Sonali Shekhawat , Maanvi Jangir

Github Repository - [link](#)